Zilionis 2019 Non-Small-Cell Lung Cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6 tumour samples (75-cell p2 dropped at the geometric-mean threshold 131 over the 232→75 spillover cliff) intermix on one continuous connected component with no per-patient islands (radius-graph: 0 detached); iLISI median 1.00→3.26 of 6. Cycling (32%) is interspersed, not a detached proliferative lobe. Read on the regenerated embedding across a θ×n_pcs sweep.

Gate 2 — module geography → GO (non-branching)

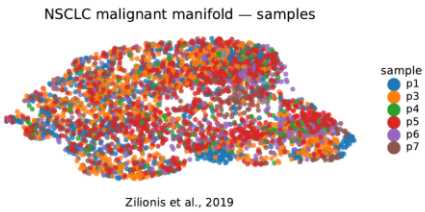
Organized Gast↔pSom cross-repression along one axis (smoothed spatial r=−0.58; Gast-high origin opposing a pSom-high end) with NO co-expression hub (Gast&pSom co-high enrichment 0.73<1). Inverse neural-crest gradient opposing both (Crest↔Gast r=−0.47, Crest↔pSom −0.37). Along pseudotime Gast falls (ρ=−0.60), pSom rises (ρ=+0.78). Crest not on a Gast arm — no flag.

Gate 3 — hub mutual exclusivity → NA

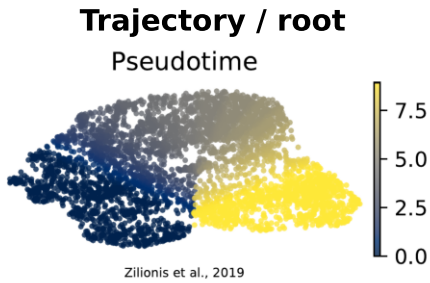
No bifurcation: the principal curve is robustly 2-tip/0-fork and fold-free across Nodes 15/20/30, while a PPT tree throws 10–20 spurious forks (a bush, not a Y), and there is no co-expression hub — so there is no branch pair to test for mutual Gast↔pSom exclusivity.

STEP FIGURES

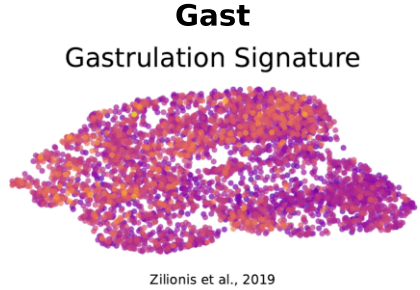
Gate 1: integration



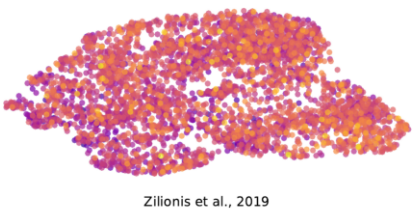
Trajectory / root



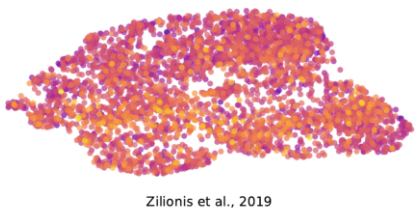
Gast



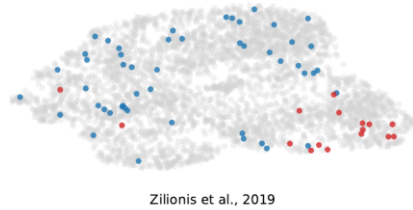
pSom



Crest



EMT-TFs



Dataset. NSCLC / non-small-cell lung cancer (Zilionis et al., 2019), inDrop — 7 samples / 31,179 cells (master scope); malignant only: 3,995 cells, of which 3,920 across 6 samples enter preprocessing (75-cell p2 dropped as spillover); the whole malignant manifold (single body, no scoping) enters the trajectory.

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Zilionis's NSCLC malignant manifold is a single continuous axis of organized Gast↔pSom cross-repression (spatial r=−0.58) rooted at the Gast/pSom-high developmental pole, with an inverse neural-crest gradient; it lacks Puram's co-expression hub-and-spoke bifurcation (Gast&pSom co-high enrichment 0.73<1; PPT bush vs a fold-free 2-tip curve), so Gate 3 has no hub to test. The compartment is 32% cycling but forms one continuous component (no separable proliferative lobe); the axis is developmental, not proliferative — the developmental modules are cell-cycle-free (Gast 0/81, Crest 0/104 cell-cycle genes) and the Gast↔pSom cross-repression survives within non-cycling cells alone (r=−0.58→−0.59), with pseudotime↔cycling near-zero (ρ=−0.14). EMT-TFs, all detected but sparse in this inDrop compartment, resolve two ends: SNAI2 (root, 5.1%) → TWIST2 (terminus, 1.7%), sparsity disclosed.

Parameter choices

Min cells/sample	√(232·75)≈131 (232→75 cliff)	**Integration**	Harmony θ=10; n_pcs=30; spread=1.3
Scope	whole manifold (single body)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = Gast/pSom-high	**Geometry**	Linear-pSom/Crest (non-branching)